

How To Select Your Own Benchtop SMD Assembly



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If you are a low-volume electronics manufacturer or a hobbyist stepping up from hand assembly, you might be overlooking how efficient you can get in production by continuing to do hand assembly. A lot of people doing just through-hole soldering might be nervous about surface mount. They might also be tweezing their way through work, which is not the most efficient way to do things. There are many things that you can do to get more efficient.

There are different flavours of benchtop or desktop printed circuit board (PCB) assembly solutions available. "Desktop SMT equipment should be compact, yet flexible enough to cover a wide spectrum of today's manufacturing needs and in the future, too. These should also be easy to use and economical," says Kumud Tyagi, vice president, Bergen Associates Pvt Ltd.

NeoDen-4



Pick-and-place

Are you ready to look at pick-and-place (PNP) machines? The most important point to keep in mind is that the least expensive machines will almost always be the most difficult to use. There is a new family of PNP machines coming out of China. A number of different products are being launched by companies who are just copying each other to come out with these machines. They lack important features and also suffer from support and service issues.

Let us take a look at some of the things that you need to consider while shopping for a PNP machine.

Select the right feeders. The components now come on paper or plastic tape. So if you are assembling manually, you would just need to hold on to the tape somehow. In PNP, you might get machines where there is no facility to advance the tape and so you will have to be careful so as not to spill the components all over the workbench. It will be a pain to sort out the small components, especially if these are polarised. The last thing you want is to rework the whole thing. Low-end machines, typically, do not have auto advance, and this means that you would have to babysit the machine while it does its job.

Machine vision. The least expensive PNP's do not have machine vision. Machine vision is a camera and software combo that works to ensure that the part is on the nozzle and aligned correctly before it puts down the component. There are some new machines from China that can be sourced from AliExpress, which do this, or you can

"Applying the paste to the board is the trickiest part of my business."

—Bob Coggeshall